

LECTURE SHEET

EXERCISE-I

(Gas Laws, Ideal gas equation)

LEVEL-I (MAIN)

Straight Objective Type Questions

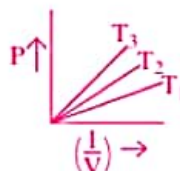
- Gases show ideal gas behaviour at
 - high pressure and high temperature
 - low pressure and high temperature
 - low pressure and low temperature
 - high pressure and low temperature
- A gas of volume 2000ml is kept in a vessel at a pressure of 10^3 pascals at a temperature of 27°C . If the pressure is increased to 10^5 pascals at the same temperature, the volume of the gas becomes
 - 1000ml
 - 20ml
 - 2ml
 - 200ml
- Which of the following indicates Charles' law mathematically (when n, P are constant)?
 - $VT = \text{constant}$
 - $V_t = V_0 \left(1 + \frac{t}{273} \right)$
 - $V_0 = V_t \left(1 + \frac{t}{273} \right)$
 - $V/T = \text{constant}$ (when n, P are constant)
 - a, c
 - a, b
 - b, c
 - b, d
- The volume of a given mass of a gas is 100ml at 100°C . If pressure is kept constant at what temperature will the sample have the volume of 200ml?
 - 50°C
 - 473°C
 - 200°C
 - 400°C

5. A gas cylinder withstands a pressure of 14.9 atm. Its pressure gauge indicates 12 atm at 27°C. If the building catches fire suddenly, at what temperature the cylinder explodes?
1) 21.7°C 2) 33.5°C 3) 372.5°C 4) 99.5°C
6. Value of R in SI units is
1) $8.315 \times 10^7 \text{ erg K}^{-1} \text{ mole}^{-1}$ 2) $8.315 \text{ J K}^{-1} \text{ mole}^{-1}$
3) $0.0823 \text{ lit. Atm. K}^{-1} \text{ mole}^{-1}$ 4) $2 \text{ cal K}^{-1} \text{ mole}^{-1}$
7. The value of the universal gas constant R depends upon the
1) Nature of the gas 2) Mass of the gas
3) Temperature of the gas 4) The units of measurement
8. Assertion (A) : 8 gm of CH₄ occupies 11.355 of volume at S.T.P.
Reason (R) : 1 mole of any gas at S.T.P. occupies 22.711 lit.
9. The molar volume of an ideal gas at one atmosphere and 273°C is
1) 22.4L 2) 44.8L 3) 11.2 L 4) 5.6 L
10. At 27°C, one mole of an ideal gas exerted a pressure of 0.821 atmospheres. What is its volume in litres ?
(R = 0.082 lit-atm/mol⁻¹ K⁻¹)
1) 300 2) 30 3) 0.3 4) 3
11. 'n' moles of an ideal gas at temperature 'T' occupy 'V' litres of volume, exerting a pressure of 'P' atmospheres. What is its concentration in mole lit⁻¹ (R = gas constant)
1) $\frac{RT}{P}$ 2) $\frac{P}{RT}$ 3) $\frac{RT}{P}$ 4) $\frac{R}{RT}$
12. Two identical vessels are filled with 44g of Hydrogen and 44g of carbon dioxide at the same temperature. If the pressure of CO₂ is 2 atm, the pressure of Hydrogen is
1) 1 atm 2) 44 atm 3) 2 atm 4) 22 atm
13. A gas occupies a volume of 300 cc at 27°C and 620 mm pressure. The volume of the gas at 47°C and 640 mm pressure is
1) 260cc 2) 310cc 3) 390cc 4) 450cc
14. If the weight of 5.6 lit of a gas at NTP is 11g, the gas may be (at.wt of P = 31, N = 14, O = 16 and Cl = 35.5)
1) PH₃ 2) COCl₂ 3) NO 4) N₂O
15. 0.3 g of a gas has a volume of 112 ml at 0°C and 2atm pressure. Its Molecular weight is
1) 60 2) 30 3) 44 4) 28
16. The vapour density of a gas is 11.2. The volume occupied by 10g of the gas at STP is
1) 10 L 2) 1 L 3) 11.2 L 4) 5.6 L
17. When the absolute temperature of a gas is doubled then the correct statements are
a) The V of a gas increases by 4 times at constant P
b) The P of a gas increases by 2 times at constant V
c) The V of a gas increases by 2 times at constant P
d) The P of a gas increases by 4 times at constant V
1) b, d 2) a, c 3) b, c 4) a, d

18. If the pressure and absolute temperature of 4 litres of SO_2 gas are doubled, the volume of this gas would be
 1) 1 litre 2) 4 litres 3) 2 litres 4) 8 litres
19. One litre of a gas weighs 2g at 300 K and 1atm pressure. If the pressure is made 0.75 atm, at what temperature will one litre of the same gas weigh 1g
 1) 600K 2) 800K 3) 900K 4) 450K
20. 'x' moles of N_2 gas at S.T.P. conditions occupy a volume of 10 litres, then the volume of '2x' moles of CH_4 at 273°C and 1.5 atm is
 1) 20 lit 2) 26.6 lit 3) 5 lit 4) 16.6 lit
21. The density of a gas is 1gm/lit at STP. At 6atm pressure and 546°C , the density of the same gas is
 1) 2 g/lit 2) 1 g/lit 3) 0.5 g/lit 4) 0.25 g/lit
22. Assertion (A) : At 273K and 0.5 atms pressure, 1L of H_2 and 1L of CO_2 contain same number of molecules.
 Reason (R) : Equal volumes of all gases contain equal number of molecules under the same conditions of temperature and pressure.

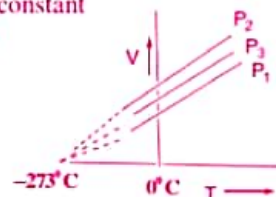
23. From the graph the correct order of temperatures is

- 1) $T_3 > T_2 > T_1$
 2) $T_3 < T_2 < T_1$
 3) $T_3 = T_2 = T_1$
 4) $T_3 > T_2 < T_1$



24. The volume-temperature graphs of a given mass of an ideal gas at constant pressures are shown below. What is the correct order of pressures?

- 1) $P_1 > P_3 > P_2$
 2) $P_1 > P_2 > P_3$
 3) $P_2 > P_3 > P_1$
 4) $P_2 > P_1 > P_3$



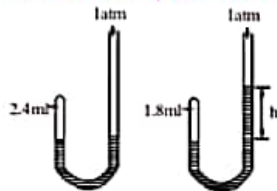
LEVEL-II (ADVANCED)

Straight Objective Type Questions

1. How much should the pressure be increased in order to decrease the volume of a gas by 5% at constant temperature
 a) 25% b) 10% c) 4.26% d) 5.26%
2. A cubical vessel has a side with 'l' cm length contained a gas at a pressure of 'P'. When the side of the vessel is made l/2 cm, the pressure of the gas becomes
 a) P b) P/8 c) 2P d) 8P
3. Four one litre flasks are separately filled with gases O_2 , F_2 , CH_4 and CO_2 under same conditions. The ratio of number of molecules in these flasks is
 a) 2:2:4:3 b) 1:1:1:1 c) 1:2:3:4 d) 2:2:3:4
4. N_2 gas is present in one litre flask at a pressure of 7.6×10^{-10} mm of Hg. The number of N_2 gas molecules in the flask at 0°C are
 a) 2.68×10^9 b) 2.68×10^{10} c) 1.34×10^{28} d) 2.68×10^{22}

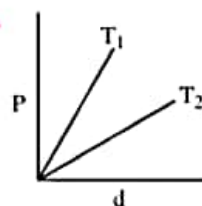
5. One mole of argon will have least density at
 a) STP b) 0°C , 2atm c) 273°C , 2atm d) 273°C , 1atm
6. By taking two J- tubes at constant temperature what is the difference in the levels of mercury in two columns ?

- a) 1013.3 mm
 b) 1140 mm
 c) 253.3 mm
 d) 760 mm

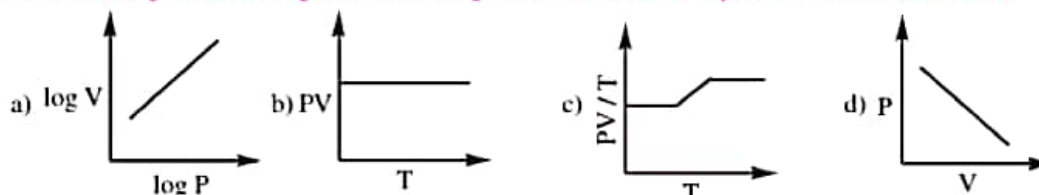


7. Diagram shows a graph between pressure and density for an ideal gas at two temperatures T_1 and T_2 which is correct

- a) $T_1 > T_2$
 b) $T = T_2$
 c) $T_1 < T_2$
 d) None



8. Which among the following indicates change in the chemical composition due to dissociation



9. Balloons of 4L capacity are to be filled with Hydrogen at a pressure of 1 atm and 27°C from an 8L cylinder containing Hydrogen at 10 atm at the same temperature. The number of balloons that can be filled is
 a) 20 b) 18 c) 40 d) 38
10. A 10 lt vessel contains He gas at 10 atm and TK. How many balloons of one litre capacity at 1 atm and 2TK can be filled by using the gas present in the cylinder
 a) 200 b) 190 c) 180 d) 170
11. An open flask has Helium gas at 2atm and 327°C . The flask is heated to 527°C at the same pressure. The fraction of original gas remaining in the flask is
 a) $3/4$ b) $1/4$ c) $1/2$ d) $2/5$

More than One correct answer Type Questions

12. The following statements are correct
 a) a plot of $\log V$ vs $\log P$ is linear at const T b) a plot of $\log V$ vs $\log T$ is linear at const P
 c) a plot of $\log P$ vs $1/T$ is linear at const V d) a plot of P vs $1/V$ is linear at const T
13. A graph is plotted between $\log V$ and $\log T$ for two moles of gas at constant pressure of 0.0821 atm. [V and T are in litres and kelvin]. Then which of the following is/are correct
 a) The graph is a straight line with slope +1 b) The graph is a straight line with slope -1
 c) The intercept on Y-axis is 0.301 d) The intercept on Y-axis is 2
14. Which of the following correctly represent the behaviour of an ideal gas
 a) $PV_m \propto T$ b) $PM \propto dT$ c) $P \propto CT$ d) $PV_m \propto CT$

15. Consider the quantity $\frac{MkT}{PV}$ of an ideal gas where M is the mass of the gas. It does not depend on the

- a) Temperature of the gas b) Volume of the gas
c) Pressure of the gas d) Nature of the gas

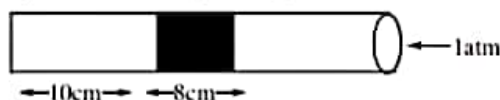
16. Boyle's law for a gas of constant mass, may be expressed as

- a) $\left(\frac{dP}{dV}\right)_T = \frac{K}{V}$ b) $\left(\frac{dP}{dV}\right)_T = -\frac{K}{V^2}$ c) $\left(\frac{dP}{dV}\right)_T = -\frac{K}{V}$ d) $V \propto \frac{1}{P}$ at constant T

Linked Comprehension Type Questions

Passage :

A 10 cm column of air is trapped by a column of mercury, 8 cm long, in a capillary tube horizontally fixed as shown below, at 1 atm pressure. When the tube is held at certain angle θ° , with open end up, the weight of Hg is borne partially by the gas. Vertical Height of Hg is a measure of additional pressure on gas.



17. The length of air column, when the tube is fixed vertically at the same temperature with open end up is

- a) $\frac{76 \times 84}{10}$ cm b) $\frac{76 \times 10}{84}$ cm c) $\frac{84 \times 10}{76}$ cm d) $\frac{76 \times 10}{68}$ cm

18. The length of air column, when the tube is fixed vertically with open end down at same temperature is

- a) $\frac{76 \times 68}{10}$ b) 0 cm c) $\frac{76 \times 10}{68}$ d) $\frac{76 \times 10}{84}$

19. When the tube is held at 45° with the horizontal with open end up the length of air column is

- a) $\frac{76 \times 10}{76 + \frac{8}{\sqrt{2}}}$ b) $\frac{76 \times \frac{8}{\sqrt{2}}}{76 + 10}$ c) $76 \times \frac{8}{\sqrt{2}}$ d) $10 \times \frac{8}{\sqrt{2}}$

Matrix Matching Type Questions

20. **Column-I**

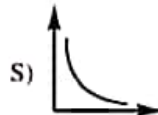
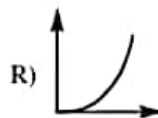
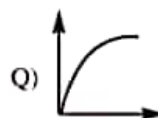
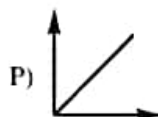
A) $\frac{1}{V^2}$ vs P for ideal gas at constant T and n.

B) V vs $\frac{1}{T}$ for ideal gas at constant P and n

C) PT vs T^2 for ideal gas at constant V and n

D) V vs $\frac{1}{P^2}$ for ideal gas at constant T and n

Column-II



Integer Type Questions

21. A gas occupies a volume of 2.5L at $9 \times 10^5 \text{ Nm}^{-2}$. The additional pressure required to decrease the volume of the gas to 1.5L keeping the temperature constant is $____ \times 10^5 \text{ Nm}^{-2}$
22. 4.0g of argon gas has pressure P and temperature TK in a vessel. On keeping the vessel at 50°C higher, 0.8g of argon was given out to maintain the pressure at P. The original temperature was $____ \times 10^2 \text{ K}$.
23. For a given mass of gas, if pressure is reduced to half and temperature is increased two times, then the volume of gas would become $____ \text{ times}$

EXERCISE-II*(Dalton's law, Diffusion of gases)***LEVEL-I (MAIN)**Straight Objective Type Questions

- Dalton's law of partial pressures is applicable to

1) $\text{NO} + \text{O}_2$	2) $\text{H}_2 + \text{Cl}_2$	3) $\text{NH}_3 + \text{HCl}$	4) $\text{CO}_2 + \text{O}_2$
-----------------------------	-------------------------------	-------------------------------	-------------------------------
- Combination that obeys Dalton's law

A = CO	B = Cl_2	C = F_2	D = Xe
1) A, B	2) B, C	3) B, D	4) A, C
- In a ten litre vessel, the total pressure of a gaseous mixture containing H_2 , N_2 and CO_2 is 9.8atm. The partial pressures of H_2 and N_2 are 3.7 and 4.2 atm respectively. Then the partial pressure of CO_2 is

1) 1.9atm	2) 0.19atm	3) 2.4atm	4) 0.019atm
-----------	------------	-----------	-------------
- O_2 and He are taken in equal weights in a vessel. The pressure exerted by Helium in the mixture is

1) $1/8^{\text{th}}$ of total pressure	2) $1/9^{\text{th}}$ of total pressure
3) $2/9^{\text{th}}$ of total pressure	4) $8/9^{\text{th}}$ of total pressure
- The total pressure of a mixture of 8g of oxygen and 14g of nitrogen contained in a 11.2L vessel at 0°C is.

1) 0.5 atm	2) 1 atm	3) 1.5 atm	4) 2 atm
------------	----------	------------	----------
- A and B are ideal gases. The molecular weights of A and B are in the ratio of 1 : 4. The pressure of a gas mixture containing equal weights of A and B is P atm. What is the partial pressure (in atm) of B in the mixture?

1) P/5	2) P/2	3) P/2.5	4) 3P/4
--------	--------	----------	---------
- A mixture contains 16g of oxygen, 28g of nitrogen and 8g of methane. Total pressure of the mixture is 740mm. What is the partial pressure of nitrogen in mm?

1) 185 mm	2) 370 mm	3) 555 mm	4) 740 mm
-----------	-----------	-----------	-----------
- At 27°C , a closed vessel contains a mixture of equal weights of helium (mol. wt = 4), methane (mol.wt = 16) and sulphur dioxide (mol. wt = 64). The pressure exerted by the mixture is 210 mm. If the partial pressure of helium methane and sulphurdioxide are P_1 , P_2 and P_3 respectively, which one of the following is correct?

1) $P_3 > P_2 > P_1$	2) $P_1 > P_2 > P_3$	3) $P_1 > P_3 > P_2$	4) $P_2 > P_3 > P_1$
----------------------	----------------------	----------------------	----------------------

9. In a gaseous mixture at 4 atm pressure, 25% of molecules are Nitrogen, 40% of molecules are carbon dioxide and the rest are oxygen. The partial pressure of oxygen in the mixture is
 1) 1.40 atm 2) 1.6 atm 3) 1 atm 4) 0.9 atm
10. 2gm of hydrogen is present in a closed vessel at S.T.P. If the same quantity of another gas 'X' when introduced into the vessel the pressure becomes 1.5 atm. The gas 'X' would be
 1) CH₄ 2) SO₂ 3) He 4) O₂
11. Mixing of two gases by diffusion is
 1) Reversible 2) irreversible 3) Exothermic 4) endothermic
12. For the diffusion of a gas at pressure P, the rate of diffusion is expressed by
 1) $r \propto \frac{P}{\sqrt{M}}$ 2) $r = \frac{P}{M}$ 3) $r \propto \frac{M}{\sqrt{P}}$ 4) $r = \frac{P}{\sqrt{M}}$
13. A balloon filled with acetylene is kept in a vessel of hydrogen at the same temperature and pressure. If the balloon is punctured with a pin, it
 1) collapses 2) bursts 3) nothing happens 4) becomes red
14. Among the following, the applications of Graham's law are
 a) In Marsh gas alarm b) In designing eudiometer
 c) In separating uranium isotopes d) To measure volume of a gas by gas burette
 1) a, c 2) b, d 3) c, d 4) b, d
15. The rate of diffusion of Hydrogen is about
 1) 1/2 that of He 2) Twice that of He 3) 1.414 times that of He 4) Four times that of He
16. Which of the pairs of gases diffuse at a slower rate than CO₂?
 1) H₂ and He 2) SO₂ and SO₃ 3) N₂ and CO 4) N₂O and C₃H₈
17. The rate of diffusion of methane is twice that of the following gas under the same conditions
 1) O₃ 2) O₂ 3) SO₂ 4) SO₃
18. At what temperature will the rate of diffusion of N₂ be 1.6 times the rate of diffusion of SO₂ at 27°C?
 1) 336°C 2) 27°C 3) 50°C 4) 63°C
19. Two samples of gases 'a' and 'b' are at the same temperature. The molecules of 'a' are travelling 4 times faster than molecules of 'b'. The ratio of M_a/M_b will be
 1) 1/4 2) 16/1 3) 4/1 4) 1/16
20. Assertion (A) : Rate of diffusion of H₂ gas is the highest.
 Reason (R) : Rate of diffusion of a gas is inversely proportional to molecular weight of the gas.
21. 350 cm³ of CH₄ and 175cm³ of an unknown gas 'A' diffused in the same time under similar conditions. The molecular mass of gas A is
 1) 32 2) 64 3) 30 4) 71
22. A certain volume of Methane diffuses in 10 Sec through a porous partition. The time taken by an equal volume of oxygen to diffuse under the same conditions is
 1) 14.14 Sec. 2) 7.07 Sec. 3) 20 sec. 4) 5 Sec
23. 0.5 mole each of H₂, SO₂ and CH₄ are kept in a container. A hole was made in the container. After 3 hours the order of partial pressures in the container will be :
 1) P_{SO₂} > P_{CH₄} > P_{H₂} 2) P_{CH₄} > P_{SO₂} > P_{H₂} 3) P_{H₂} > P_{SO₂} > P_{CH₄} 4) P_{H₂} > P_{CH₄} > P_{SO₂}

24. 50 ml of oxygen diffuses under certain conditions through a porous membrane. The volume of Hydrogen that diffuses in the same time under the same conditions is
- 1) 12.5 ml 2) 25ml 3) 100ml 4) 200ml

LEVEL-II (ADVANCED)

Straight Objective Type Questions

- In a mixture of N_2 and CO_2 gases, the partial pressure of CO_2 is 1.25 atm. The total pressure of the mixture is 5atm. The mole fraction of N_2 in the mixture is
a) 0.82 b) 0.75 c) 0.80 d) 0.65
- Equal volumes of two jars contain HCl, NH_3 gases respectively at constant temperature and pressure P. When one of the jars is inverted over another jar so that they mix up, the pressure in either of the jars is
a) 1 atm b) Equal to P c) Becomes Zero d) $P + P = 2P$
- When 2g of a gas A is introduced into an evacuated flask kept at $25^\circ C$, the pressure is found to be 1 atmosphere. If 3g of another gas is then added to the same flask, the total pressure becomes 1.5 atm. Assuming ideal behaviour, the ratio of their molecular weights $M_A : M_B$ is
a) 3 : 1 b) 1 : 3 c) 2 : 3 d) 3 : 2
- A vessel has N_2 gas saturated with water vapour at a total pressure of 1 atm. The partial pressure of water vapour is 0.3 atm. The contents of this vessel are completely transferred to another vessel having one third of the capacity of the original volume, at the same temperature. The total pressure of this system in the new vessel is
a) 3.0 atm b) 1 atm c) 3.33 atm d) 2.4 atm
- The reaction between gaseous NH_3 and HBr produces a white solid NH_4Br . Suppose that NH_3 and HBr are introduced simultaneously into the opposite ends of an open tube of 1 metre length. Where would you expect the white solid to form?
a) At a distance of 34.45 cm from NH_3 end b) At a distance of 68.5 cm from NH_3 end
c) At a distance of 44.45 cm from HBr end d) At a distance of 45.45 cm from HBr end
- Two glass bulbs A and B are connected by very small tube having a stop cock. Bulb A has a volume of 100 ml and contained a gas while bulb B is empty and has a volume of 150 ml. On opening the stop cock, the pressure of the gas in bulb A will fall down to :
a) 80% b) 60% c) 40% d) 20%
- 10 lts of an air sample with relative humidity 0.6 is compressed to 5 litres at same temperature. What is the partial pressure of the water vapour in the compressed air, if the pressure of saturated water vapour at that temperature is 2.4 K Pa
a) 1.2 KPa b) 2.4 Kpa c) 3.6 KPa d) 1.8 KPa
- Two containers A and B have the same volume. Container 'A' contains 5 mole oxygen gas. Container 'B' contains 3 moles He gas and 2 mole CH_4 gas. Both the containers are separately kept in vacuum at the same temperature. Both the containers have very small orifices of the same area through which the gases leak out. The rate of diffusion of O_2 is how many times the rate of diffusion of He and CH_4 mixture
a) $\sqrt{\frac{8.8}{32}}$ b) $\sqrt{\frac{7}{32}}$ c) $\frac{3}{2}\sqrt{\frac{8.8}{32}}$ d) $\sqrt{\frac{32}{8.8}}$

More than One correct answer Type Questions

9. Two vessels are connected by a valve of negligible volume. One container, A has 2.8 g N_2 at $T_1 K$. The other container, B is completely evacuated. The container A is heated to $T_2 K$, while container B is maintained at $\frac{T_2}{3} K$. Volume of A is half that of B. If the valve is opened, then
- 6/70 moles of N_2 will be present in B
 - The weight ratio of N_2 in A and B vessels is 1:6
 - N_2 gas does not enter into B vessel
 - number of moles of N_2 released into B vessel is less than that in A vessel
10. To measure the pressure exerted by a trapped gas A, an open ended mercury manometer is used. After sparking the gas A, it dissociates as, $A_{(g)} \rightarrow B_{(g)} + 3C_{(g)}$. If the pressure of A decreases to 0.9 atm, then (assume temperature to be constant, 300K)
- total pressure increased to 1.3 atm
 - total pressure increased by 0.3 atm
 - difference in mercury level is 228 mm
 - Total pressure increased by 0.3 cm of Hg
11. Under identical conditions of pressure and temperature
- $\frac{r_{H_2}}{r_{O_2}} = 4$
 - $\frac{r_{H_2}}{r_{CH_4}} = 2.83$
 - $\frac{r_{He}}{r_{CH_4}} = 2$
 - $\frac{r_{He}}{r_{SO_2}} = 4$

Matrix Matching Type Questions**12. Column-I**

- Boyle's law
- Avagadro's law
- Charles' law
- Dalton's law

Column-II

- $P_{\text{obs}} = P_{\text{atm}} + P_{\text{water vapour}}$
- $V_1 = n_1 \left(\frac{V_2}{n_2} \right)$
- $V_1 = V_0 \left(1 + \frac{t}{273} \right)$
- $V_1 = P_2 \left(\frac{V_2}{P_1} \right)$

Integer Type Questions

- In what ratio by mass carbon monoxide and nitrogen should be mixed so that partial pressure exerted by each gas is same ?
- 3.2g of oxygen (At.wt =16) and 0.2g of hydrogen (At.wt = 1) are placed in a 1.12 litre flask at $0^\circ C$. The total pressure of the gas mixture will be _____ atm
- A vessel contains equal weights of He and CH_4 gases at 20 bar pressure. Due to the leakage, the gases in the vessel started effusing out. What is the volume ratio of He and CH_4 gases coming out initially
- Two gram of hydrogen diffuse from a container in 10 minutes. How many grams of oxygen would diffuse through the same container in the same time under similar conditions?

EXERCISE-III

(Kinetic theory, molecular velocities, Collision Parameters)

LEVEL-I (MAIN)

Straight Objective Type Questions

- The kinetic gas equation is applicable when the gas is present in a
 - Cubic vessel
 - Spherical vessel
 - Vessel of any shape
 - Cylindrical vessel
- The correct statements regarding kinetic molecular theory are
 - The distance between the molecules is high compared to size of the gaseous molecules
 - The motion of the gaseous molecules are affected by gravitational force
 - The attractive forces between the gaseous molecules are very high.
 - The total K.E of a sample of gaseous molecules remains constant at a given temperature
 - b, d
 - b, c
 - a, d
 - c, d
- Which is wrong according to kinetic theory
 - Average K.E. of molecules is proportional to the absolute temperature
 - Collisions between molecules are perfectly elastic
 - Pressure is due to collisions between molecules
 - There are no attractive forces between the molecules of a gas
- When gas molecules collide on the walls of the vessel, the energy of the molecules changes into
 - Heat
 - Temperature
 - Light
 - None
- The Kinetic energy of a gas depends upon
 - nature of the gas
 - absolute temperature
 - molecular weight of the gas
 - number of moles of the gas
 - b, d
 - c, d
 - a, d
 - a, b
- Assertion (A) : At a given temperature 1 mole of any gas has same kinetic energy.
Reason (R) : Kinetic energy = $(3/2)RT$ (for 1mole of gas).
- The ratio of kinetic energies of 2gm of H_2 and 4gm of CH_4 at a given temperature is
 - 4:1
 - 2:32
 - 1:4
 - 16:2
- The kinetic energy of 1 mole of oxygen molecules in cal mol^{-1} at 27°C
 - 300
 - 600
 - 900
 - 800
- The kinetic energy of 'N' molecules of H_2 is 3J at -73°C . The kinetic energy of the same sample of H_2 at 127°C is
 - 12 J
 - 6 J
 - 9 J
 - 3 J
- Which of the following indicates RMS velocity of a gas ?
 - $\sqrt{\frac{2RT}{M}}$
 - $\sqrt{\frac{3P}{d}}$
 - $\sqrt{\frac{8RT}{\pi M}}$
 - $\sqrt{\frac{3RT}{M}}$
 - c, d
 - b, d
 - a, c
 - a, b

11. Which of the following is incorrect ?
 1) RMS velocity depends upon molecular weight
 2) RMS velocity depends upon temperature
 3) RMS velocity depends upon density at a given temperature
 4) The RMS velocity is used in deriving the kinetic gas equation
12. When pressure is increased upon a gas at constant temperature
 1) the R.M.S. velocity decreases
 2) the R.M.S. velocity increases
 3) the R.M.S. velocity remains the same
 4) the average kinetic energy of the molecules increases
13. Which of the following relations regarding molecular velocities are true ?
 a) Most probable velocity = $0.8166 \times \text{RMS velocity}$
 b) RMS velocity = $0.9213 \times \text{Average velocity}$
 c) Average velocity = $\sqrt{\frac{8RT}{\pi M}}$
 d) $C_p > \bar{C} > C$
 1) b, c 2) a, d 3) c, d 4) a, c
14. The root mean square velocity of a gas is doubled when the temperature is
 1) reduced to one-half 2) reduced to one-fourth
 3) increase four times 4) increased two times
15. Assertion (A) : H_2 and O_2 have same R.M.S. velocity at the same temperature.
 Reason (R) : R.M.S. velocity of a gas molecules is directly proportional to square root its 'T'.
16. At the same temperature carbon monoxide molecules have the same most probable velocity as the molecules of
 1) Nitrogen dioxide 2) Nitrogen 3) Nitrous Oxide 4) Oxygen
17. R.M.S velocity of ethane molecules when one mole of ethane is present in 10 lt vessel at 10 atm is
 1) 10^2 cm/sec 2) 10^3 cm/sec 3) 10^4 cm/sec 4) 10^5 cm/sec
18. R.M.S velocity of helium at S.T.P is $x \text{ cm/sec}$. R.M.S velocity of helium at 273 K and 4 atm is
 1) $4x \text{ cm/sec}$ 2) $3x \text{ cm/sec}$ 3) $2x \text{ cm/sec}$ 4) $x \text{ cm/sec}$
19. RMS velocity of a gas at a given temperature is how much factor higher than its most probable velocity
 1) $\frac{3}{2}$ 2) $\frac{\sqrt{3}}{2}$ 3) $\sqrt{\frac{3}{2}}$ 4) $2/3$
20. The most probable velocity of the molecules of a gas is 1 km/sec . The R.M.S velocity of the molecules is
 1) 1.128 km/sec 2) 1.224 km/sec 3) 1.5 km/sec 4) 1.086 km/sec
21. If two gases of molecular weights M_A and M_B at temperature T_A and T_B are taken such that, $T_A M_B = T_B M_A$, then which property has the same magnitude for both the gases.
 1) density 2) pressure 3) KE per mol 4) u_{rms}

LEVEL-II (ADVANCED)

Straight Objective Type Questions

- At what temperature the most probable velocity of O_2 gas is equal to the RMS velocity of O_3 at $t^\circ C$?
 a) $(273 + t) \sqrt{\frac{3}{2}} K$ b) $(273 + t) \sqrt{\frac{2}{3}} K$ c) $(273 + t) K$ d) $\frac{3}{2} (273 + t) K$
- What is the RMS speed of a moist particle of mass $10^{-12} g$ at $27^\circ C$ according to kinetic molecular theory?
 a) 0.35 cm/sec b) 0.45 cm/sec c) 0.6 cm/sec d) 0.7 cm/sec
- The rate of effusion of two gases 'a' and 'b' under identical conditions of temperature and pressure are in the ratio of 2:1. What is the ratio of rms velocity of their molecules if T_a and T_b are in the ratio of 2:1?
 a) 2 : 1 b) $\sqrt{2} : 1$ c) $2\sqrt{2} : 1$ d) $1 : \sqrt{2}$
- Incorrect statement among the following is
 a) Larger the size of molecule, smaller is the mean free path
 b) Greater the number of molecules per unit volume, smaller is the mean free path
 c) Larger the temperature, larger is the mean free path
 d) Larger the pressure, larger is the mean free path
- Collision frequency (Z) of a gas at a particular pressure
 a) decreases with the rise in temperature b) increases with the rise in temperature
 c) decreases initially and thereafter increases d) unpredictable.
- The closest distance between the centres of two molecules of a gas taking part in collision is called
 a) molecular diameter b) collision diameter c) both (a) and (b) d) none of these

More than One correct answer Type Questions

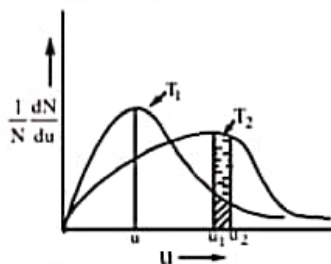
- If an inert gas expands at constant pressure by providing heat
 a) The temperature increases
 b) The intermolecular attractive forces decrease
 c) The kinetic energy of molecules decrease
 d) The number of molecules per unit volume increases
- According to kinetic theory of gases, for a diatomic molecule, which is not correct
 a) the pressure exerted by the gas is proportional to the mean speed of the molecule
 b) the pressure exerted by the gas is proportional to the mean square speed of the molecule
 c) the root mean square speed of the molecule is inversely proportional to the square root of temperature
 d) the mean translational kinetic energy of the molecule is proportional to the absolute temperature
- Which one of the following is true about the effect of increase in temperature on the distribution of molecular speeds in a gas?
 a) the area under the distribution curve remains the same as that under lower temperature
 b) the distribution becomes broader
 c) the fraction of the molecules with most probable speed increases
 d) the most probable speed of the molecules increases
- For a mixture of 1 mole He and 1 mole Ne, select the correct statements(s)
 a) Molecules of the two gases strike the wall of the container with same frequency
 b) Molecules of the helium strike the wall more frequently
 c) $u_{av(He)} > u_{av(Ne)}$ d) $P_{He} > P_{Ne}$

11. There are equal volumes of N_2 and CO_2 gases at $25^\circ C$ and 1 atm. Then indicate the correct statements among the following
- The rms velocity remains the same for N_2 and CO_2 gases
 - The average translational K.E per molecule is the same for N_2 and CO_2
 - The density of N_2 is less than that of CO_2
 - The total translational K.E of both N_2 and CO_2 is the same
12. If the u_{rms} of an ideal gas in a closed container is doubled, then which of the following statement correctly explains how the change is accomplished?
- The temperature is doubled
 - The pressure is quadrupled at constant volume
 - The temperature is quadrupled
 - The pressure is quadrupled at constant temperature
13. Mean free path decreases if
- H_2 is replaced by He
 - He is replaced by H_2
 - He is replaced by O_2
 - O_2 is replaced by CO_2

Linked Comprehension Type Questions

Passage-I :

According to Maxwell distribution, the area under the curve is equal to the total number of molecules in collision. On increasing temperature, fraction of molecules having speed equal to u_{mp} (most probable speed) decreases. The speed distribution also depends on the mass of the molecules along with the temperature. In general, the distribution depends upon the value of $\frac{M}{T}$ (where M is molar mass and T is temperature in kelvin).



14. Relationship between T_1 and T_2 is given as
- $T_1 = T_2$
 - $T_1 > T_2$
 - $T_2 > T_1$
 - can't be determined
15. On increasing the temperature, number of molecules having speed between u_1 and u_2
- increases
 - decreases
 - remains constant
 - can't be determined
16. If u_{mp} at $T_1 K$ is u , then value of u_{rms} at T_2 is
- $3\sqrt{\frac{T_2}{T_1}}u$
 - $\sqrt{\frac{3T_2}{2T_1}}u$
 - $\sqrt{\frac{T_2}{T_1}}u$
 - $2\sqrt{\frac{T_2}{T_1}}u$

Passage-II :

The root mean square speed of an ideal gas is given by : $u_{rms} = \sqrt{\frac{3RT}{M}}$. Thus we conclude that u_{rms} speed of the ideal gas molecules is proportional to square root of the temperature and inversely proportional to the square root of the molar mass. The translational kinetic energy per mole can also be given as $\frac{1}{2}Mu_{rms}^2$. The mean free path (λ) is the average of distances travelled by molecules in between two successive collisions whereas collision frequency (C.F.) is expressed as number of collisions taking place in unit time. The two terms λ and C.F. are related by : $C.F. = \frac{u_{rms}}{\lambda}$

17. A jar contains He and H_2 in the molar ratio 1 : 5. The ratio of mean translational kinetic energy at the same temperature is
 a) 1 : 5 b) 5 : 1 c) 2 : 1 d) 1 : 1
18. Which of the following relation is correct for an ideal gas regarding its pressure (P) and translational kinetic energy per unit volume (E) ?
 a) $P = \frac{2}{3}E$ b) $P = \frac{3}{2}E$ c) $P = \frac{1}{2}E$ d) $P = 2E$
19. If n represents number of moles, n_0 is number of molecules per unit volume, k is Boltzmann constant, R is molar gas constant, T is absolute temperature and N_A is Avogadro's number then which of the following relations is wrong ?
 a) $P = n_0 k T$ b) $P = n_0 R T$ c) $P = \frac{n K N_A T}{V}$ d) $n_0 = N_A \times \frac{n}{V}$

Matrix Matching Type Questions20. **Column-I****Column-II**

- | | |
|---|---|
| A) If temperature of given gas is increased | P) Average speed of gas molecules will increase |
| B) If the pressure of the given gas increased at constant temperature | Q) Root mean square speed of gas molecule will increase |
| C) If the density of a given gas is lowered at constant temperature | R) Most probable speed of gas molecule will increase |
| D) If the volume of a given gas is increased at constant temperature | S) Speed of gas molecules will not change |

Integer Type Questions

21. The K.E of N molecules of O_2 is x Joules at $-123^\circ C$. Another sample of O_2 at $27^\circ C$ has a KE of 2x Joules. The latter sample contains _____ N molecules of O_2 .
22. At $27^\circ C$, the average speed of N_2 molecules is $x \text{ ms}^{-1}$. At 927° , the speed will be _____ $x \text{ ms}^{-1}$
23. The mass of molecule A is twice the mass of molecule B. The rms speed of A is twice the rms speed of B. If two samples of A and B contain same number of molecules, what will be the ratio of pressures of two samples in separate containers of equal volume. _____
24. Collision diameter is least in case of (1) H_2 ; (2) He ; (3) CO_2 ; (4) N_2 . Indicate with the number in brackets.

EXERCISE-IV*(Real gases, Van der waal's equation, Critical condition, Joule-Thomson effect)***LEVEL-I (MAIN)****Straight Objective Type Questions**

1. The deviation from the ideal gas behaviour of a gas can be expressed as

1) $Z = \frac{P}{VRT}$

2) $Z = \frac{PV}{nRT}$

3) $Z = \frac{nRT}{PV}$

4) $Z = \frac{VR}{PT}$

2. Which of the following are correct statements ?
 X) Vander Waal's constant a is a measure of attractive force
 Y) Vander Waal's constant b is also called co-volume or excluded volume
 Z) b is expressed in L mol^{-1}
 1) X, Y 2) Y, Z 3) X, Z 4) X, Y, Z
3. In Vander Waal's equation of state of the gas law, the constant ' b ' is a measure of
 1) Intermolecular repulsions 2) Intermolecular collisions per unit volume
 3) Volume occupied by the molecules 4) Intermolecular attraction
4. For ammonia vander waals parameter ' a ' value is $4.17 \text{ lit}^2 \cdot \text{atm mol}^{-2}$. For NH_3 , ' b ' value is
 1) 5.2 mol/lit 2) 5.2 lit/mol 3) 0.037 lit/mol 4) 0.037 mol/lit
5. Compressibility factor for H_2 behaving as real gas is
 1) 1 2) $\left(1 - \frac{a}{RTV}\right)$ 3) $\left(1 + \frac{Pb}{RT}\right)$ 4) $\frac{RTV}{(1-a)}$
6. An ideal gas cannot be liquified because
 1) collisions are elastic 2) of the absence of attractive forces
 3) gravity has no effect on the molecules
 4) the average K.E. is not proportional to the absolute temperature
7. The critical temperature of a substance is defined as
 1) the temperature above which the substance decomposes
 2) the temperature above which a substance can exist only as a gas
 3) melting point of the substance 4) boiling point of the substance
- 8.

Gas	O_2	N_2	NH_3	CH_4
a	1.360	1.390	4.170	2.253

The table indicates the value of vander Waal's constant a in $\text{L}^2 \text{atm mol}^{-2}$. The gas which can most easily be liquefied is ?

- 1) O_2 2) N_2 3) NH_3 4) CH_4
9. Which set of conditions represents easiest way to liquefy a gas ?
 1) Low temperature and high pressure 2) High temperature and low pressure
 3) Low temperature and low pressure 4) High temperature and high pressure
10. The gas which can be liquified under high pressure at 4°C is
 1) nitrogen 2) hydrogen 3) oxygen 4) ammonia
11. The vander Waal's parameters of two gases are given as
- | | $a (\text{dm}^6 \text{bar mol}^{-2})$ | $b (\text{dm}^3 \text{mol}^{-1})$ |
|-------|---------------------------------------|-----------------------------------|
| Gas A | 6.5 | 0.056 |
| Gas B | 18.0 | 0.011 |
- Considering the value of parameters, which of the following statement(s) is/are correct?
 1) Critical volume of A < critical volume of B
 2) Critical pressure of A > critical pressure of B
 3) Critical temperature of A < critical temperature of B
 4) Ease of liquification of A > ease of liquification of B

12. The relationship between P_c , V_c and T_c is
 1) $P_c V_c = RT$ 2) $P_c V_c = 3RT_c$ 3) $P_c V_c = \frac{3}{5} RT_c$ 4) $P_c V_c = \frac{3}{8} RT_c$
13. The Joule Thomson coefficient is zero at
 1) Inversion temperature 2) Critical temperature
 3) Absolute temperature 4) Below 0°C
14. A gas X causes heating effect when allowed to expand. This is because
 1) The gas is a noble gas 2) The inversion temperature of the gas is very low
 3) The gas is ideal gas 4) The boiling point of the gas is very low
15. The gas that is heated up during Joule Thomson effect at ordinary temperature is
 1) O_2 2) CO_2 3) H_2 4) SO_2

LEVEL-II (ADVANCED)

Straight Objective Type Questions

1. Positive deviation from ideal behaviour takes place because of
 a) Molecular interaction b/w atoms and $\frac{PV}{nRT} > 1$ b) Molecular interaction b/w atoms and $\frac{PV}{nRT} < 1$
 c) Finite size of atoms and $\frac{PV}{nRT} > 1$ d) Finite size of atoms and $\frac{PV}{nRT} < 1$
2. Choose the wrong statement from among the following.
 a) The corrected pressure $P + \frac{an^2}{V^2}$ in Vander Waals equation (symbols have their usual meanings) is the pressure which the gas would exert if it were ideal
 b) Above their respective Boyle's temperatures, H_2 shows positive deviation throughout ($z > 1$) while O_2 shows negative deviation ($z < 1$) followed by positive deviation ($z > 1$)
 c) A gas shows negative deviation when long range attractive intermolecular forces are dominating
 d) The intercept of PV vs P isotherm of any gas at 27°C is equal to $24.4 \text{ L atm mol}^{-1}$
3. At 273 K temperature and 9 atm pressure, the compressibility for a gas is 0.9 . The volume of 1 milli-moles of gas at this temperature and pressure is :
 a) 2.24 litre b) 0.020 mL c) 2.24 mL d) 22.4 mL
4. 44 g of CO_2 gas is present in 10 litre vessel at TK . The pressure exerted by CO_2 is how many torrs less than the pressure exerted by ideal gas ($a = 3.6 \text{ lt}^2 \cdot \text{atm} \cdot \text{mole}^{-2}$, $b = 0.043 \text{ lt/mol}$)
 a) 0.036 b) 27.4 c) 0.36 d) 274
5. 2 moles of Xenon is present in a one litre vessel. For Xenon gas $a = 4.19 \text{ lt}^2 \cdot \text{atm} \cdot \text{mol}^{-2}$, $b = 0.05 \text{ lt/mol}$. The volume of free space available for random motion of molecules is
 a) 900 c.c b) 100 c.c c) 950 c.c d) 50 c.c

More than One correct answer Type Questions

6. Which of the following are the characteristics of a real gas?
 a) The molecules attract each other
 b) It shows deviations from the ideal gas law
 c) It obeys the gas law at low temperature and high pressure
 d) The mass of molecules is negligible

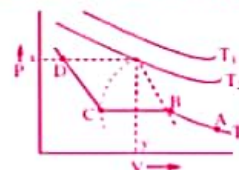
7. Which of the following gases deviate from ideal behaviour to a greater extent
 a) H_2 b) NH_3 c) He d) SO_2
8. A gas obeys $P(V-b) = RT$. Then which of the following are correct
 a) Isochoric curves have the slope = $R/V-b$
 b) Isobaric curves have the slope = R/P with intercept = b
 c) For the gas compressibility factor = $1 + Pb/RT$
 d) Attractive forces predominate repulsive forces
9. Select the correct statements
 a) At high pressure real gases are less compressible in comparison to ideal gas
 b) H_2 and He are more compressible than ideal gases at all pressures
 c) Except H_2 and He, the compressibility factor is less than unity for all gases at low pressure
 d) The compressibility factor of real gases is independent of temperature
10. When a sample of a gas is compressed at constant temperature from 15 atm to 60 atm; its volume changes from 76 cm^3 to 20.5 cm^3 . Which of the following are possible.
 a) The gas shows non-ideal behaviour b) The gas dimerizes
 c) The gas is adsorbed onto the vessel walls d) The gas molecules have repulsions to each other
11. Which of the following are correct
 a) For hydrogen gas $Z > 1$, at room temperature and pressure
 b) The real gas behaves as ideal gas at Boyle's temperature
 c) For Vander waal's gas $T_C = \frac{8a}{27Rb}$
 d) At low pressure, for a Vander waal's gas, $Z = 1 + \frac{Pb}{RT}$
12. Vander Waal's constants for three different gases are given :
- | Gas | a | b |
|-----|------|-------|
| X | 3.0 | 0.025 |
| Y | 10.0 | 0.030 |
| Z | 6.0 | 0.035 |
- Which is correct ?
 a) maximum critical temperature – Y
 b) most ideal behaviour – X
 c) maximum molecular volume – Z
 d) minimum molecular volume – X
13. Which of the following statements is/are correct about the coefficient B in the virial equation of state

$$PV_m = RT \left(1 + \frac{B}{V_m} + \frac{C}{V_m^2} + \dots \right)$$

 a) It is independent of temperature
 b) It is equal to zero at Boyle temperature
 c) It has dimensions of molar volume
 d) It is dependent on pressure

14. Isotherms of CO_2 gas at different temperatures are given below ($a = 3.6 \text{ L}^2 \text{ atm. mol}^{-2}$, $b = 0.043 \text{ L/mol}$). Correct statement(s) among the following is/are

- Product of X and Y is equal to $9.3 \text{ atm lit.mol}^{-1}$
- T_1 may be 293 K
- T_3 may be 350 K
- From state B to C liquid CO_2 exist



Linked Comprehension Type Questions

Passage-I :

Gases tend to behave non-ideally at low temperatures and high pressures. The deviation from ideal behaviour can be explained by considering two types of corrections. They are volume correction and pressure correction.

15. Select incorrect statement(s) :

- Volume correction is due to finite size of molecules and pressure correction is due to force of attraction between molecules.
- At high temperatures, molecules have greater kinetic energy, and attractive forces are smaller and the behaviour of gases is close to the ideal gas behaviour
- Volume correction is also called covolume or excluded volume and is four times the volume of spherical molecules present in one mole of the gas
- At very low pressure, force of attraction is effective and pressure correction needs further resolution.

16. Following represents equation of state for n moles of real gas

$$\left[P + \frac{n^2 a}{V^2} \right] [V - nb] = nRT \quad \text{Select incorrect statement for a real gas}$$

- Constant 'a' is a measure of force of attraction among gas molecules
- a is expressed in $\text{atm L}^2 \text{ mol}^{-2}$, b is expressed in L mol^{-1}
- At high pressure, compression factor is $\left(1 + \frac{Pb}{RT} \right)$
- $\frac{n^2 a}{V^2}$ is also called internal volume

17. Which assumption of kinetic theory is not followed when a real gas shows non-ideal behaviour ?

- Gas molecules move at random with no attractive forces between them
- The velocity of gas molecules is dependent on temperature
- Amount of space occupied by a gas is much greater than that by actual gas molecules
- During collisions with the walls of the container or with another molecules, energy is conserved

Passage-II :

Ideal gas equation is represented as $PV = nRT$. Gases present in the universe were found ideal in the

Boyle's temperature range only. The compressibility factor for an ideal gas, $Z = \frac{PV}{nRT} = 1$. The main cause to show deviations were due to wrong assumptions made about forces of attractions (which becomes significant at high pressure). Volume occupied by molecules V_i in $PV = nRT$, is supposed to be volume of gas or, the volume of container in which gas is placed by assuming that gaseous molecules do not have appreciable volume. Actual volume of the gas is that volume in which each molecule of a gas can move freely. If volume occupied by gaseous molecule is not negligible, then the term V should be replaced by the ideal volume which is available for free motion of each molecule of gas.

Similarly for n moles of gas $V_{\text{actual}} = V - nb$

The excluded volume can be calculated by considering bimolecular collisions. The excluded volume is the volume occupied by the sphere of $2r$ for each pair of molecule.

Thus, excluded volume for one pair of molecules $= \frac{4}{3}\pi(2r)^3 = \frac{4 \times 8\pi r^3}{3}$

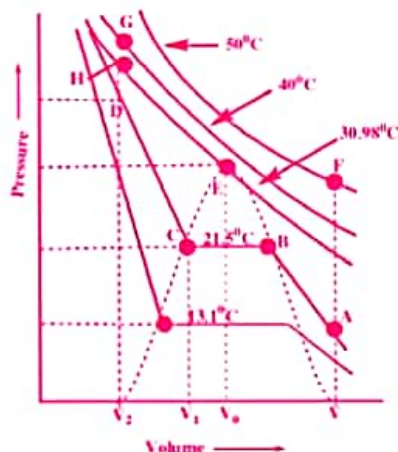
The excluded volume for 1 molecule $= \frac{2}{3} \times 8\pi r^3 = 4 \times \left(\frac{4}{3}\pi r^3\right) = 4V$

The excluded volume for N molecules $= 4NV = b$ (where N is Avogadro's No.)

18. Assuming the molecules of gas as hard spheres of radius 2.0×10^{-10} m the fraction of volume occupied by the molecules to the total volume of a given amount of gas at 27°C and at 1 bar pressure and 10 bar pressure respectively are :
- a) 99.9%, 99% b) 0.082%, 0.82% c) 99%, 90% d) 11%, 10%
19. The compressibility factor for N_2 at -50°C and 800 atm pressure is 1.95. Number of moles of N_2 required to fill a balloon of 100 L. capacity is :
- a) 2.244×10^3 b) 2.244×10^2 c) 2.244 d) 22.44

Passage-III :

Study the following isotherms of carbondioxide at various temperature and answer the following questions based on that



20. Choose the temperature above which CO_2 cannot be liquified whatsoever applied pressure is
- a) 290K b) 294.5 K c) 302K d) 304 K
21. At 21.5°C decrease in volume(B-C) doesn't result into increase in pressure because
- a) The compression causes increase in solidification
b) At point B only all gases convert into solid
c) The compression causes further increase in condensation
d) Carbon dioxide is real gas
22. Carbondioxide present at F can be liquified
- a) At critical volume by compression b) At critical pressure by compression
c) By compression at any temperature d) By compression and lowering the temperature

Matrix Matching Type Questions23. **Column-I**

- A) Hydrogen gas ($P = 200 \text{ atm}$, $T = 273\text{K}$)
 B) Hydrogen gas ($P = 0$, $T = 273\text{K}$)
 C) CO_2 ($P = 1 \text{ atm}$, $T = 273\text{K}$)
 D) Real gas with very large molar volume

Column-II

- P) Compressibility factor $\neq 1$
 Q) Attractive forces are dominant
 R) $PV = nRT$
 S) $P(V - nb) = nRT$

24. Match the compression factor under different condition (in Column-I) with its value (in Column-II):

Column-I

- A) Compression factor (Z) for ideal gas
 B) Z for real gas at low P
 C) Z for real gas at high P
 D) Z for critical state

Column-II

- P) $\frac{3}{8}$
 Q) $(1 + Pb/RT)$
 R) 1
 S) $(1 - a/RTV)$

25. **Column-I**

- A) Atm L mol^{-1}
 B) $\sqrt{\frac{2RT}{M}}$
 C) Boyle temperature
 D) Volume of one mole

Column-II

- P) a/Rb
 Q) a/b
 R) V_m
 S) u_{MP}

Integer Type Questions

26. Excluded volume for free random motion of gas molecules is how many times the actual volume of gas molecules
 27. Inversion temperature of a gas is how many times the Boyle's temperature?
 28. What is the value of $\sqrt[3]{\frac{a}{P_c b^2}}$?
 29. The volume to be excluded due to only two molecules of a gas in collision with a fixed point of impact is $\frac{0.09}{N_A}$ litre ($N_A = \text{Avogadro number}$). If the value of ' a ' is $3.6 \text{ atm L}^2\text{mol}^{-2}$, then the value of Boyle's temperature is, 10^xK . What is the value of ' x '? ($R = 0.08 \text{ L atm K}^{-1}\text{mol}^{-1}$)

EXERCISE-V*(Intermolecular forces, Vapour pressure, Viscosity, Surface tension)***LEVEL-I (MAIN)**Straight Objective Type Questions1. F_2 is gas but I_2 is solid, because

- 1) Larger london forces are present in I_2 when compared to F_2
- 2) Lesser number of london forces are present in I_2 when compared to F_2
- 3) F_2 and I_2 has same extent of london forces
- 4) I_2 has low bond dissociation energy

2. Among the following molecules, London or dispersion forces are strongest in
1) n-hexane 2) n-pentane 3) isopentane 4) neopentane
3. Non polar compounds can also solidify because of
1) Van der waals forces 2) Dipole – dipole interaction
3) Ionic bonds 4) Hydrogen bonds
4. Which forces of attraction are responsible for liquification of hydrogen chloride
1) coulombic forces 2) dipole forces
3) hydrogen bonding 4) vander Waal's forces
5. The inter molecular forces present in inert gases are
1) Ion – ion 2) Ion – dipole
3) Dipole – dipole 4) Dispersion
6. Regarding dipole - dipole attractions the incorrect statement is
1) Dipole - dipole attractions are more if the molecules have high dipole moment values.
2) In liquid HBr, dipole - dipole attractions are present.
3) Dipole - Dipole interaction energy between stationary polar molecules $\propto r^{-3}$.
4) Dipole - dipole interaction energy between rotating molecules $\propto r^{-6}$.
7. In ion-dipole forces, the magnitude of the interaction energy (E)
1) $E = \frac{Z^2\mu}{r^2}$ 2) $E = \frac{Z\mu}{r}$ 3) $E = \frac{Z\mu^2}{r^2}$ 4) $E = \frac{Z\mu}{r^2}$
8. Hydration of different ions is an example of
1) Ion - dipole interaction 2) Dipole - dipole interaction
3) Dipole - induced dipole 4) Dispersion
9. If thermal energy predominates over intermolecular forces, then the substance changes from ___to ___
1) gas to liquid 2) liquid to solid
3) gas to solid 4) liquid to gas
10. A liquid is in equilibrium with its vapour at its boiling point. On average, the molecules in the two phases have equal
1) Potential energy 2) Kinetic energy 3) Total energy 4) 1, 2 and 3
11. A relation between vapour pressure and temperature is known as
1) Ideal gas equation 2) Boltzmann equation
3) Clausius equation 4) Clausius - Clapeyron equation
12. Find incorrect match
1) Unit of surface energy = $J-m^{-2}$
2) Unit of surface tension = $N - m^{-1}$
3) Molecules on the surface of liquid = less energy.
4) Minimum surface area of a liquid = Lowest energy state
13. The surface tension of water at 20°C is 72.75 dyne cm^{-1} . Its value in SI system is
1) 7.275 $N m^{-1}$ 2) 0.7275 $N m^{-1}$ 3) 0.07275 $N m^{-1}$ 4) None of the above

14. A manifestation of surface tension is :
1) Rise of liquid in a capillary tube
2) Spherical shape of liquid drops
3) Upward movement of water in soils
4) All the above
15. On heating a liquid, its surface tension
1) Increases
2) Decreases
3) Remains same
4) Is reduced to zero
16. Generally liquid drops assume spherical shape because :
1) A sphere has maximum surface area
2) A sphere has minimum surface area
3) Sphere is symmetrical in shape
4) Sphere is heavy
17. Laminar flow of a liquid means
1) Regular gradation of velocity for layers in passing from one layer to the next layer of a liquid
2) Showing constancy in the velocity of layers of a liquid
3) Increase in the velocity of layers from surface to bottom of a liquid
4) All
18. Find incorrect statement
1) Due to viscosity, velocity of flow of water at the surface is more than that at the bottom in a river.
2) Due to viscosity, velocity of flow of water at the surface is less than that at the bottom in a river
3) Velocity gradient = du/dx
4) Viscosity coefficient is related to absolute temperature as $\eta = Ae^{E/RT}$
19. In SI system, the units of coefficient of viscosity, η are
1) $\text{Kg s}^{-1}\text{m}^{-2}$
2) $\text{Kg m}^{-1} \text{s}^{-1}$
3) $\text{Kg cm}^{-1} \text{s}^{-1}$
4) $\text{g m}^{-1} \text{s}^{-1}$
20. Poise stands for
1) $1 \text{ dynes cm sec}^{-2}$
2) $1 \text{ dyne sec cm}^{-2}$
3) 10^{18} e.s.u-cm
4) 10^{-7} erg sec.
21. The viscosity of four liquids P, Q, R and S are 8.5, 11.4, 18 and 12.3 respectively, then which flows slowly.
1) P
2) Q
3) R
4) S
22. With rise in temperature of a liquid, the viscosity
1) Increases
2) Decreases
3) Remains constant
4) May increase or decrease
23. The thickness of window panes of old buildings is more at the bottom than at the top, which is due to
1) Surface tension of glass
2) Viscosity of glass
3) Expansion of solid at a given temperature
4) Expansion of liquid at a given temperature
24. Among the following compounds viscosity is highest for
1) Methanol
2) Propane-1,2,3 triol
3) Ethyleneglycol
4) Ethanol

KEY SHEET (LECTURE SHEET)

EXERCISE-I

LEVEL-I

- 1) 2 2) 2 3) 4 4) 2 5) 4 6) 2 7) 4 8) 1
 9) 2 10) 2 11) 2 12) 2 13) 2 14) 4 15) 2 16) 1
 17) 3 18) 2 19) 4 20) 2 21) 1 22) 1 23) 1 24) 1

LEVEL-II

- 1) d 2) d 3) b 4) b 5) d 6) c 7) a 8) c
 9) b 10) c 11) a 12) abd 13) ac 14) abc 15) abc 16) bd
 17) b 18) c 19) a 20) A-R; B-S; C-P; D-Q 21) 6 22) 2
 23) 4

EXERCISE-II

LEVEL-I

- 1) 4 2) 3 3) 1 4) 4 5) 3 6) 1 7) 2 8) 2
 9) 1 10) 3 11) 2 12) 1 13) 2 14) 2 15) 3 16) 2
 17) 3 18) 4 19) 4 20) 3 21) 2 22) 1 23) 1 24) 4

LEVEL-II

- 1) b 2) c 3) b 4) d 5) b 6) c 7) b 8) a
 9) ab 10) abc 11) abcd 12) A-S; B-Q; C-R; D-P 13) 1 14) 4
 15) 8 16) 8

EXERCISE-III

LEVEL-I

- 1) 3 2) 3 3) 3 4) 4 5) 1 6) 1 7) 1 8) 3
 9) 2 10) 2 11) 3 12) 3 13) 4 14) 3 15) 4 16) 2
 17) 4 18) 4 19) 3 20) 2 21) 4

LEVEL-II

- 1) c 2) a 3) c 4) d 5) b 6) b 7) ab 8) ac
 9) abd 10) ac 11) bcd 12) bc 13) bcd 14) c 15) a 16) b
 17) d 18) a 19) b 20) A-PQR; B-S; C-S; D-S 21) 1 22) 2
 23) 8 24) 2

EXERCISE-IV

LEVEL-I

- 1) 2 2) 4 3) 3 4) 3 5) 3 6) 2 7) 2 8) 3
 9) 1 10) 4 11) 3 12) 4 13) 1 14) 2 15) 3

OBJECTIVE CHEMISTRY IB**STATES OF MATTER****LEVEL-II**

- 1) c 2) b 3) c 4) b 5) a 6) ab 7) bd 8) abc
9) ac 10) ad 11) abc 12) abcd 13) bc 14) abcd 15) d 16) d
17) a 18) b 19) a 20) d 21) c 22) d
23) A-PS; B-R; C-PQ; D-R 24) A-R; B-S; C-Q; D-P
25) A-Q; B-S; C-P; D-R 26) 4 27) 2 28) 3 29) 3

EXERCISE-V**LEVEL-I**

- 1) 1 2) 1 3) 1 4) 2 5) 4 6) 4 7) 4 8) 1
9) 4 10) 2 11) 4 12) 3 13) 3 14) 4 15) 2 16) 2
17) 2 18) 2 19) 2 20) 2 21) 3 22) 2 23) 2 24) 2